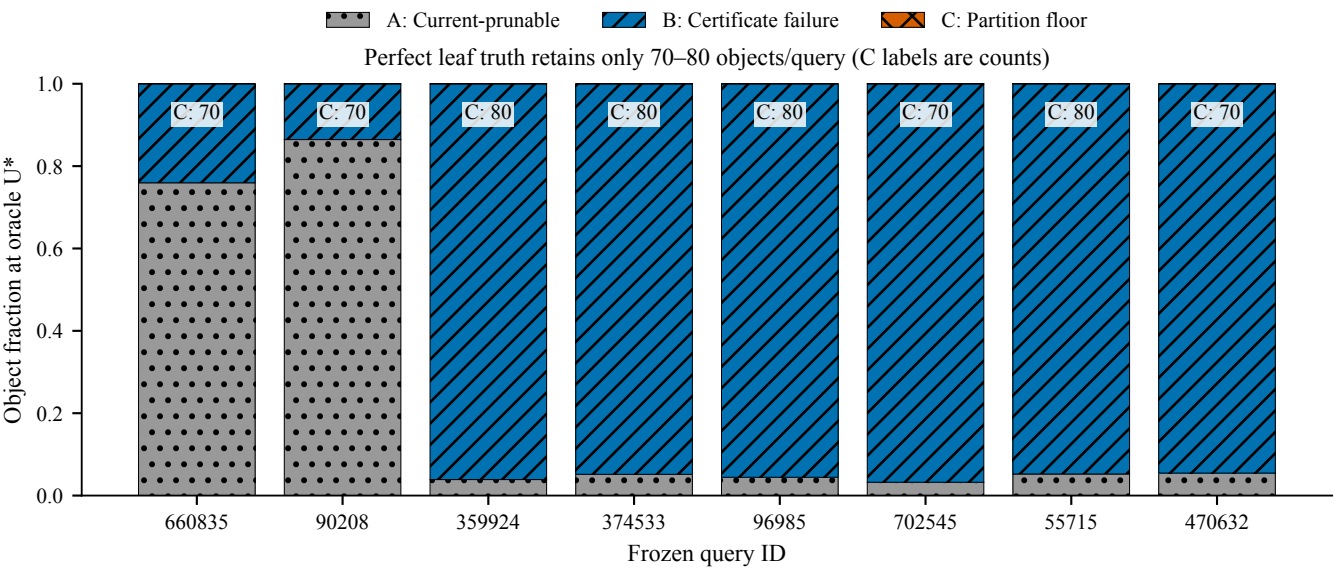
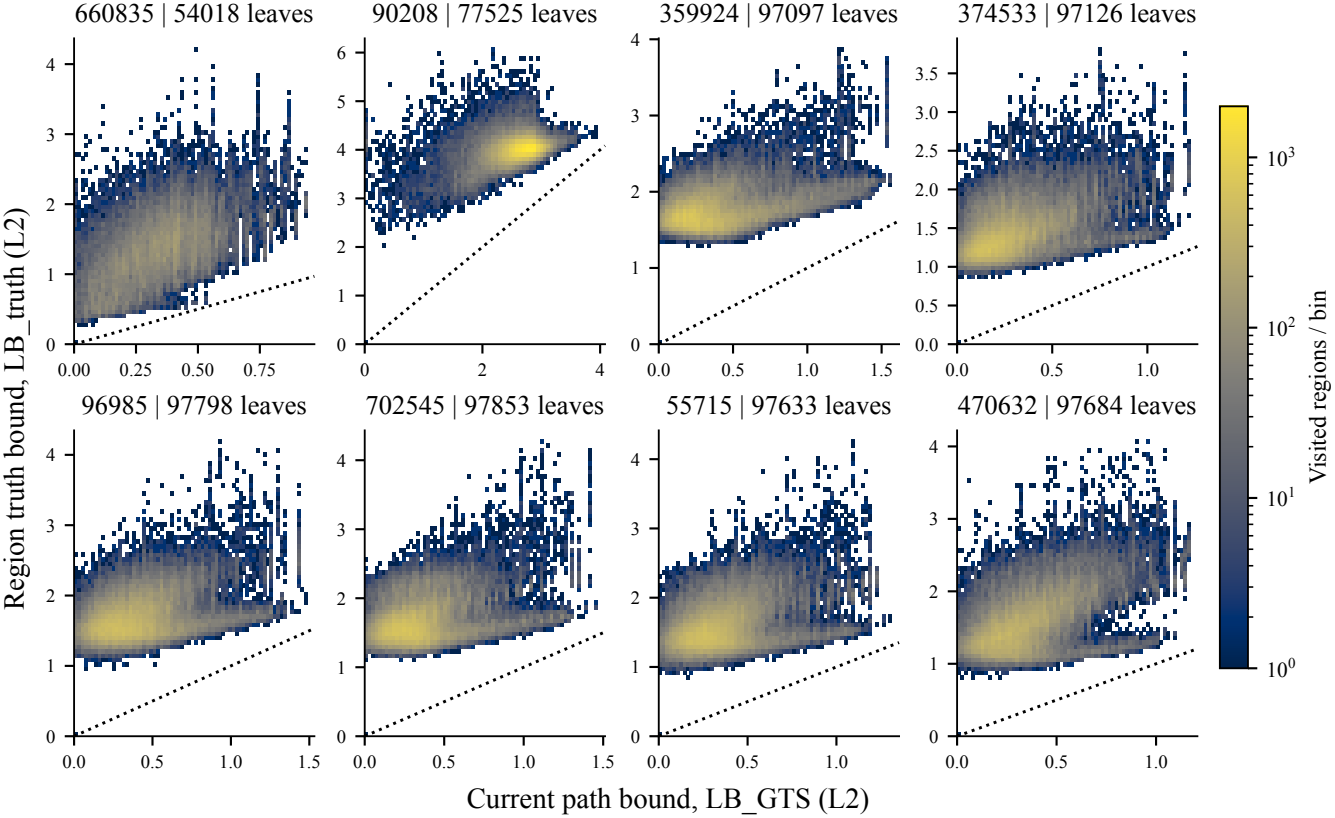
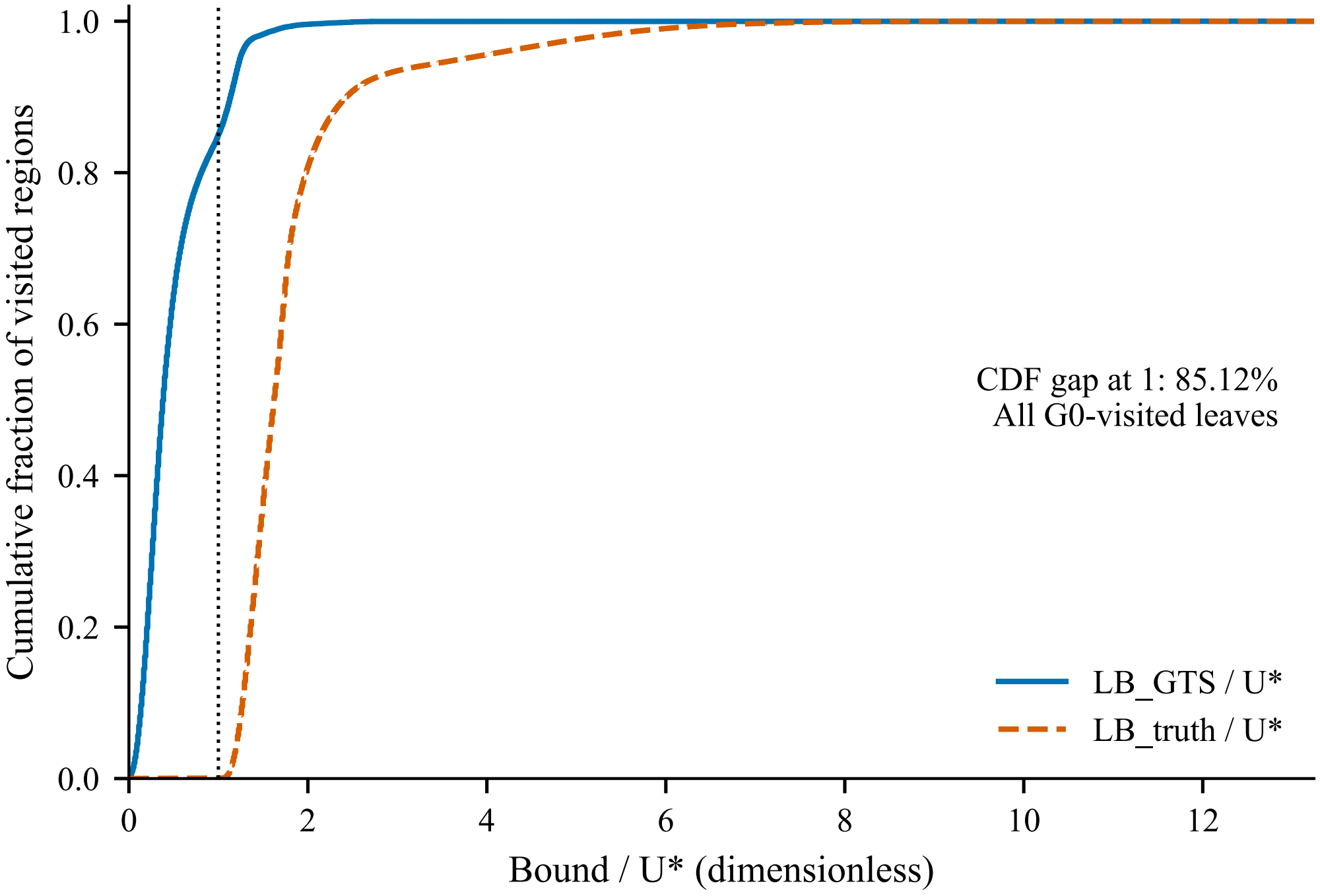
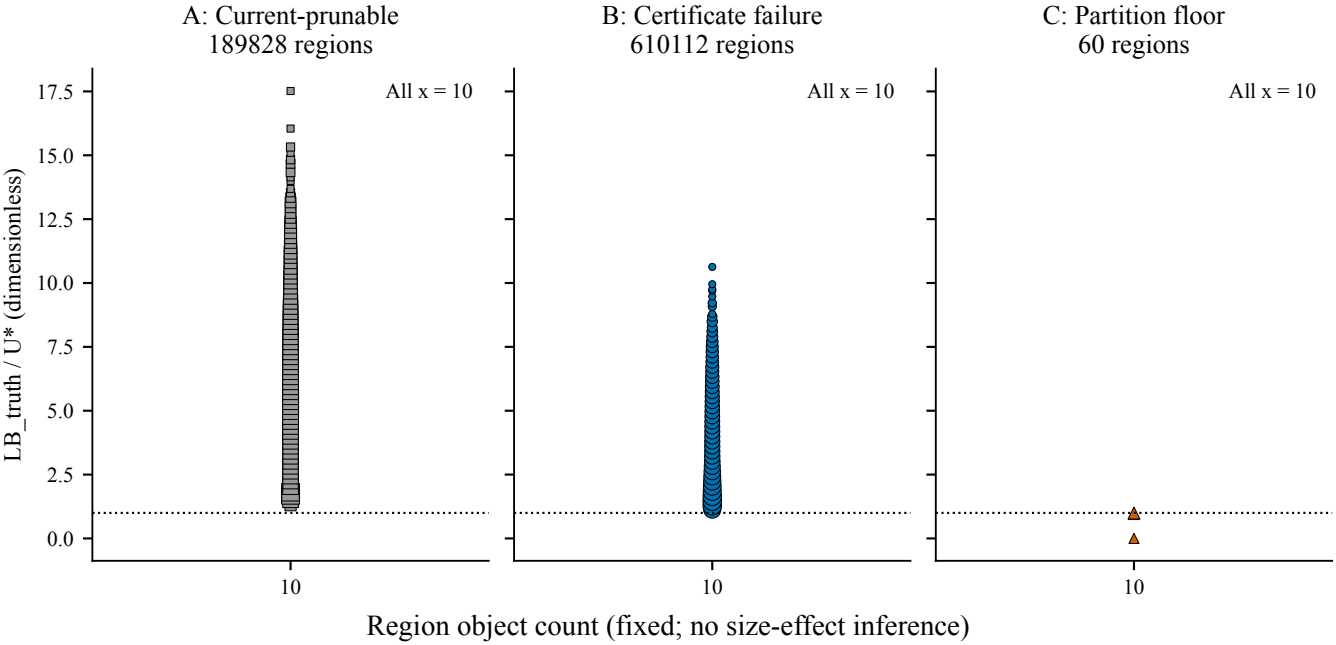

A: Current-prunable
B: Certificate failure
C: Partition floor
Perfect leaf truth retains only 70–80 objects/query (C labels are counts)
Object fraction at oracle U*
1.0
0.8
0.6
0.4
0.2
0.0
C: 70
C: 70
C: 80
C: 80
C: 80
C: 70
C: 80
C: 70
660835
90208
359924
374533
96985
702545
55715
470632
Frozen query ID


660835 | 54018 leaves
90208 | 77525 leaves
359924 | 97097 leaves
374533 | 97126 leaves
96985 | 97798 leaves
702545 | 97853 leaves
55715 | 97633 leaves
470632 | 97684 leaves
Region truth bound, LB_truth (L2)
Current path bound, LB_GTS (L2)
Visited regions / bin
10^0
10^1
10^2
10^3


1.0
0.8
0.6
0.4
0.2
0.0
Cumulative fraction of visited regions
0
2
4
6
8
10
12
Bound / U* (dimensionless)
CDF gap at 1: 85.12%
All G0-visited leaves
LB_GTS / U*
LB_truth / U*


A: Current-prunable
189828 regions
B: Certificate failure
610112 regions
C: Partition floor
60 regions
All x = 10
All x = 10
All x = 10
LB_truth / U* (dimensionless)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
10
10
10
Region object count (fixed; no size-effect inference)